

# Multipattern graph convolutional network-based autism spectrum disorder identification

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The early diagnosis of autism spectrum disorder (ASD) has been extensively facilitated through the utilization of resting-state fMRI (rs-fMRI). With rs-fMRI, the functional brain network (FBN) has gained much attention in diagnosing ASD. As a promising strategy, graph convolutional networks (GCN) provide an attractive approach to simultaneously extract FBN features and facilitate ASD identification, thus replacing the manual feature extraction from FBN. Previous GCN studies primarily emphasized the exploration of topological simultaneously connection weights of the estimated FBNs while only focusing on the single connection pattern. However, this approach fails to exploit the potential complementary information offered by different connection patterns of FBNs, thereby inherently limiting the performance. To enhance the diagnostic performance, we propose a multipattern graph convolution network (MPGCN) that integrates multiple connection patterns to improve the accuracy of ASD diagnosis. As an initial endeavor, we endeavored to integrate information from multiple connection patterns by incorporating multiple graph convolution modules. The effectiveness of the MPGCN approach is evaluated by analyzing rs-fMRI scans from a cohort of 92 subjects sourced from the publicly accessible Autism Brain Imaging Data Exchange database. Notably, the experiment demonstrates that our model achieves an accuracy of 91.1% and an area under ROC curve score of 0.9742. The implementation codes are available at <https://github.com/immutablejackz/MPGCN>.

**Key words:** resting-state fMRI; autism spectrum disorder; multipattern; brain connectivity networks; graph convolution network.

## Introduction

Autism spectrum disorder (ASD) is a neurodevelopmental condition that is characterized by a typical brain development that significantly impacts an individual's social cognition and interaction abilities (Lord et al. 2020). As a major and growing global health issue, ASD impacts an estimated 40–50 million people worldwide (Wen et al. 2022). The current diagnosis of ASD, which relies on behavioral observations and disease symptoms (Muskett et al. 2019), heavily relies on the expertise of clinicians (Karampasi et al. 2020), thereby limiting opportunities for optimal treatment. Consequently, there has been a growing focus within the fields of psychiatry and neuroscience on the pursuit of an objective biomarker that can facilitate an early detection and prompt intervention in the management of autism.

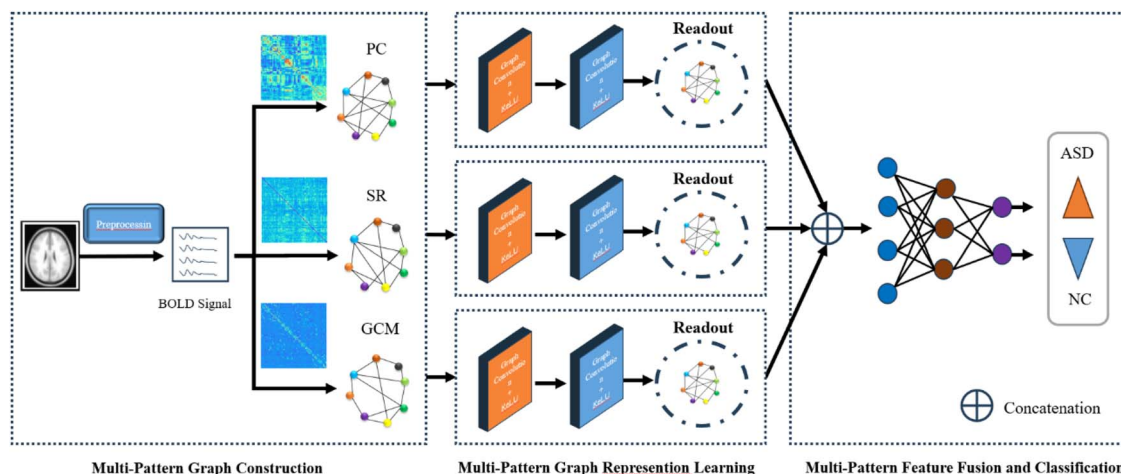
In recent years, there has been a particular emphasis on the integration of fMRI with deep learning techniques for the diagnosis of ASD (Haghighat et al. 2022; Shao et al. 2023). This integration allows for a more effective understanding of the underlying causes, thereby improving the diagnosis and treatment of neurological disorders (Zhang et al. 2020). However, relying solely on the fMRI data for identification poses challenges due to the random nature of brain activity and the asynchrony in timing among subjects (Li et al. 2017; Yang et al. 2023).

The use of resting-state fMRI (rs-fMRI) to construct and analyze the functional brain networks (FBNs) has brought great promise

to the classification of brain disorders. By considering each region of interest (ROI) as a node and the strength of connectivity as edges, FBN enables the identification of both topological variations and differences in the connection strength between normal and disrupted states induced by pathological factors (Li et al. 2019; Zhang et al. 2022). These variations serve as potential biomarkers to comprehend the underlying pathological mechanisms of ASD (Peng et al. 2022). However, literature research has indicated that constructing FBNs networks for ASD prediction is a complex undertaking, requiring the acquisition of network features (Wang et al. 2019) and relying heavily on specialized knowledge (Karampasi et al. 2020). The graph neural network (GNN) is a potent deep learning technique that is capable of autonomously acquiring graph representations in a data-centric fashion (Wu et al. 2020, 2022). The development of GNNs has consequently emerged as a powerful approach to effectively discern fMRI biomarkers within FBNs, with the ultimate goal of expediting disease diagnosis (Lu and Uddin 2023). Essentially, each brain network can be perceived as an intricate graph structure encompassing unstructured data. These data include not only the characteristics of individual nodes but also the interconnections or topology information among different nodes (Zhang M et al. 2018). Current research has demonstrated that the utilization of graph convolutional networks (GCNs), as a comprehensive deep learning framework, facilitates the automatic capture of both

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**Fig. 1.** The entire process of multiple pattern networks for ASD diagnosis.

node characteristics and the topological relationships between nodes (Saeidi et al. 2022). By leveraging advanced techniques, such as convolution and graph pooling, GCNs facilitate the analysis of fMRI data, enabling the automatic extraction of high-level topological information from brain networks. The diagnostic accuracy of brain diseases is significantly enhanced as a consequence (Ma et al. 2023). In order to facilitate the analysis of functional connectivity (FC) and computational simulation of human brain functions, several studies have already partitioned the brain into multiple ROIs, subsequently employing GCN models to extract features and diagnose diseases in the FC networks (Zhang X et al. 2018; Meng and Ge 2022). The previous studies have primarily concentrated on examining the concurrent topological characteristics and connection weights of estimated FBNs based on a predefined connectivity pattern. However, this approach falls short in uncovering the potential complementary information present in FBNs across different connection patterns.

In order to address this issue, we propose a multipattern graph convolution network (MPGCN) for ASD diagnosis based on rs-fMRI. As illustrated in Fig. 1, we initially construct multipattern graphs for each participant, where each graph corresponds to distinct FBNs. The procedure involves initially parcellating the brain into 116 ROIs utilizing an automated anatomical labeling (AAL) atlas (Rolls et al. 2020). Subsequently, we employ 3 connectivity modes to build the brain networks, i.e. Pearson's correlation (PC), sparse representation (SR), and Granger causality mapping (GCM). Then, we propose to employ multiple GCN modules to acquire multipattern graph representations for each subject, which are subsequently fused together to incorporate multipattern characteristics. Next, the fused features are propagated through 3 fully connected layers, and the classification is performed based on the output of the final fully connected layer, enabling the disease diagnosis. To validate the efficacy of the MPGCN method, we performed experiments on a cohort of 92 subjects utilizing the rs-fMRI data sourced from the Autism Brain Imaging Data Exchange (ABIDE) database. The comparison between the combination of partial connectivity modes and individual connectivity modes demonstrated a significant enhancement in the accuracy of ASD diagnosis through our MPGCN approach.

## Related works

### Multipattern learning

Multipattern learning is a widely studied machine learning approach that deals with instances that have multiple feature

patterns (Xu et al. 2013). The motivation behind this learning mechanism is primarily driven by the data characteristics originating from real-world applications, where instances are delineated by diverse sets of features or distinct "views." For example, in multimedia content, a multimedia clip can encompass its video and audio signals simultaneously. In addition to that, multiview techniques have broad applications in medicine. Matthias et al. proposed a model called joint independent component analysis to examine simultaneous EEG–fMRI measurements from multiple participants within a single experimental session (Moosmann et al. 2008). Wang et al. have recently started combining PET/MRI scans and fMRI in the surgical series of mTLE-HS patients to detect abnormal outcomes and have achieved promising predictive results (Wang et al. 2022). Kang et al. constructed 2 datasets based on the ABIDE dataset and employed 3 methods to extract brain network features for ensemble learning. They achieved an improved classification performance compared to single-pattern learning.

### Graph convolutional networks

With the advancement of deep learning techniques, GCNs have been increasingly employed in the diagnosis and classification of certain diseases. Gu et al. utilized GCN models to extract features from brain connectivity networks for classifying different stages of Alzheimer's disease (Gu et al. 2021). Shao et al. proposed an ASD classification method that combines deep feature selection (DFS) and GCNs. They achieved an improved classification performance for ASD by utilizing DFS in conjunction with GCNs (Shao et al. 2021). By combining with multiview techniques, the classification performance of GCNs has been significantly improved. Zhang et al. employed a multiview GCN to differentiate Parkinson's disease cases from control cases. Compared to single-view techniques, the multiview approach greatly enhanced the classification accuracy (Zhang X et al. 2018; Li et al. 2021).

## Materials and methods

### Data acquisition

In this paper, we utilize the identical dataset employed in a recent investigation (Wee et al. 2016). The dataset consists of the rs-fMRI data obtained from 45 individuals with ASD and 47 typically developing individuals (referred to as NC subjects) aged between 7 and 15 years. All of the data utilized in this study are openly accessible at the ABIDE database (Di Martino et al. 2014).

**Table 1.** Demographic information of the subjects.

|                   | ASD (N = 45)             | NC (N = 47) | P-value             |
|-------------------|--------------------------|-------------|---------------------|
| Gender (M/F)      | 36/9                     | 36/11       | 0.2135 <sup>a</sup> |
| Age (year ± SD)   | 11.1 ± 2.3               | 11.0 ± 2.3  | 0.7773 <sup>b</sup> |
| FIQ (mean ± SD)   | 106.8 ± 17.4             | 13.3 ± 14.1 | 0.0510              |
| ADI-R (mean ± SD) | 32.2 ± 14.3 <sup>c</sup> | —           | —                   |
| ADOS (mean ± SD)  | 13.7 ± 5.0               | —           | —                   |

ASD: autism spectrum disorder; NC: normal control; FIQ: full intelligence quotient; ADI-R: autism diagnostic interview-revised; ADOS: autism diagnostic observation schedule. <sup>a</sup>The P-value in our analysis was calculated using the chi-squared test. <sup>b</sup>The P-value in our analysis was calculated using the 2-sample 2-tailed t-test. <sup>c</sup>Two patients do not have the ADI-R score.

Table 1 presents an overview of the demographic details pertaining to the individuals encompassed within the dataset. The diagnosis of ASD was determined based on the autism criteria outlined in the Diagnostic and Statistical Manual of Mental Disorders, 4th Edition, Text Revision (DSM-IV-TR). In order to evaluate the psychopathology for distinguishing diagnoses and identifying any coexisting Axis-I disorders, an interview was conducted either with a parent or with the individual themselves. For individuals aged below 17.9 years, the parent's interview was based on the Schedule of Affective Disorders and Schizophrenia for Children-Present and Lifetime Version (KSADS-PL). For individuals aged 18.0 years and above, the participant's interview involved the use of the Structured Clinical Interview for DSM-IV-TR Axis-I Disorders, Nonpatient Edition (SCID-I/NP) along with the Adult ADHD Clinical Diagnostic Scale (ACDS).

To qualify as NC subjects, individuals underwent interviews using KSADS-PL, SCID-I/NP, and ACDS to rule out any current Axis-I disorders. Additionally, comorbid ADHD was excluded by ensuring that all DSM-IV-TR criteria for ADHD were met except for criterion E.

## Data preprocessing

The rs-fMRI images were obtained using a standard echo-planar imaging sequence on a 3T Siemens Allegra scanner commonly used in clinical practice. Participants were instructed to relax and focus their gaze on a white fixation cross displayed against a black background projected onto a screen throughout the 6-min scanning session. The imaging parameters included a flip angle of 90°, acquisition of 33 slices, repetition time of 2,000 ms, echo time of 15 ms, collection of 180 volumes, and voxel thickness set at 4.0 mm. To preprocess the data, we utilized the widely employed statistical parametric mapping software (SPM8). To minimize the signal artifacts, the first 10 volumes of each subject were excluded during preprocessing of the rs-fMRI images. The remaining images underwent the following calibration steps: (i) standardization to the MNI space at a resolution of  $3 \times 3 \times 3 \text{ mm}^3$ ; (ii) nuisance signal regression, including ventricle, white matter, global signals, and head motion, using Friston's 24-parameter model (Friston et al. 1996); (iii) application of band-pass filtering within the frequency range of 0.01–0.08 Hz; and (iv) signal detrending. Subsequently, the preprocessed BOLD time series signals were partitioned into 116 ROIs based on the AAL atlas. Finally, these time series were organized into a data matrix, denoted as  $X \in \mathbb{R}^{170 \times 116}$ .

## Proposed method

In this section, we will present our proposed MPGCN method, comprising 3 main components: (i) multipattern FBNs construction using AAL atlases with 116 ROIs, (ii) multipattern graph

representations for each subject, and (iii) ASD diagnosis through the fusion and classification of multipattern features.

### Multipattern graph construction

Consider an undirected graph  $G = \{V, E, A\}$ , where  $V$  represents the set of nodes or ROIs with  $n$  elements,  $E$  represents the collection of edges, and  $A$  is an adjacency matrix in  $\mathbb{R}^{n \times n}$  that characterizes the interconnections among the nodes/ROIs in a specific FBN.

In this study, we constructed brain networks based on the AAL atlas using 3 connectivity patterns. The 3 connectivity patterns are as follows:

### Pearson's correlation

In every FC matrix, the connectivity between 2 ROIs is indicative of the Pearson's correlation (PC) coefficient calculated from their respective mean time series signals. The weight of the connection  $w_{ij}$  linking the  $i$ th and  $j$ th ROIs can be defined as follows:

$$w_{ij} = \frac{(x_i - \bar{x}_i)^T (x_j - \bar{x}_j)}{\sqrt{(x_i - \bar{x}_i)^T (x_i - \bar{x}_i)} \sqrt{(x_j - \bar{x}_j)^T (x_j - \bar{x}_j)}}, \quad (1)$$

where we define  $x_i \in \mathbb{R}^t$  as a representation of a sequence of BOLD signals derived from the  $i$ th ROI;  $t$  represents the total count of time points or steps;  $\bar{x}_i \in \mathbb{R}^t$  signifies an average value calculated from all elements within vector  $x_i$ ;  $i = 1, 2, \dots, n$ ; and  $x_i - \bar{x}_i$  is a centralized counterpart of  $x_i$ .

### Sparse representation

Unlike PC, which quantifies the overall correlation between variables, SR is a commonly used method for modeling partial correlation, as described by Li et al. (2019). The SR model is represented as follows:

$$\min_w \sum_{i=1}^n \left\| x_i - \sum_{j \neq i} w_{ij} x_j \right\|^2 + \lambda \sum_{j \neq i} |w_{ij}|, \quad (2)$$

where  $X = [x_1, x_2, \dots, x_n] \in \mathbb{R}^{t \times n}$  denotes the fMRI data matrix for a specific subject. Each individual brain region is associated with a specific time course, which is represented by each column of  $X$ .

### Granger causality mapping

Granger causality mapping (GCM) is a method used to model the effective connectivity, which captures the causal relationships among nodes in a network. Unlike symmetric connections, the connections in GCM are nonsymmetric. Specifically, when provided with 2 time series,  $x[n]$  and  $y[n]$ , the Granger causality

mapping process from  $x[n]$  to  $y[n]$  is defined as follows:

$$F_{x,y} = \ln \frac{|\sum (\zeta_i)|}{|\sum (\eta_i)|}, \quad (3)$$

where  $\zeta_i$  and  $\eta_i$  correspond to the residuals obtained from the regression models with restrictions and without restrictions, respectively. The symbol  $\Sigma$  denotes the variance.

We measure the connectivity between the current ROI and other ROIs to determine the edges of the FBNs. It is assumed that the edge weights are positive, which is in accordance with the theory of structural equilibrium (Heider 1946; Cartwright and Harary 1956). This assumption ensures a balanced estimated network structure when all edge weights are positive. Additionally, this presumption facilitates subsequent analysis of FBN and convolution operations by considering that numerous BC indicators, including mutual information (Salvador et al. 2007), exhibit nonnegativity. The marginal weight ( $e_{ij}$ ) is used to measure the magnitude of marginal connections between brain regions, encompassing both positive facilitation and negative inhibition. Then, the  $n$ -dimensional vector ( $|e_{i1}|, |e_{i2}|, \dots, |e_{in}|$ ),  $n = 116$  can be utilized to depict the level of connectivity between the  $i$ th brain region and its counterparts. Therefore, we define the adjacency matrices  $A_{PC}^{(116)} \in \mathbb{R}^{116 \times 116}$ ,  $A_{SR}^{(116)} \in \mathbb{R}^{116 \times 116}$ , and  $A_{GCM}^{(116)} \in \mathbb{R}^{116 \times 116}$  based on 3 connectivity modes. Simultaneously, we define  $A \in \{A_{PC}^{(116)}, A_{SR}^{(116)}, A_{GCM}^{(116)}\}$ . Ultimately, 3 connectivity patterns are utilized to construct 3 graphs (i.e.  $G_{PC} = G(X_{PC}, A_{PC})$ ,  $G_{SR} = G(X_{SR}, A_{SR})$  and  $G_{GCM} = G(X_{GCM}, A_{GCM})$ ) for each subject.

### Learning graph representations by incorporating multiple patterns

In this section, we consider brain FBNs as irregular graphs with internal structures. To analyze these FBNs and acquire novel graph representations for ASD diagnosis, we employ spectral GCNs following the approach introduced by Zhang et al. (Zhang M et al. 2018). The spectral GCN leverages the Fourier transform and its inverse counterpart to enable an efficient information aggregation among nodes in the spectral realm following the approach proposed by Bruna et al. (2013) and Kawahara et al. (2017).

The Laplacian matrix ( $L$ ) is utilized in the graph's Fourier transform and can be defined as  $L = D - A$ . In this equation, each diagonal element  $D_{ij}$  of the diagonal matrix  $D$  represents the degree of the  $i$ th node. To be more precise,  $D_{ij}$  is determined by adding up the weights of all edges that are linked to the  $i$ th node, i.e.  $D_{ii} = \sum_j A_{ij}$ . A commonly encountered variant of the Laplacian matrix is the normalized symmetric Laplacian matrix:

$$L = D^{-1/2} L D^{-1/2} = I_n - D^{-1/2} A D^{-1/2}, \quad (4)$$

where  $I_n$  is an identity matrix.

The decomposition of the Laplacian matrix  $L$  can be expressed as  $U \Lambda U^T$ , where  $U = \{u_i\}_{i=1}^n$  represents the orthogonal eigenvectors,  $\Lambda = \text{diag}(\{\alpha_i\}_{i=1}^n)$  is a diagonal matrix, and  $\alpha_i$  represents the eigenvalues associated with  $u_i$ . The matrix  $U$  allows for transforming variables into the spectral domain, thereby enabling convolutional operations on the graph (Kipf and Welling 2016).

The multiplication of the Fourier transforms of signals can be expressed as the multiplication of the Fourier transforms of the signals, as shown in the work by Shuman et al. (2013). In the context of graph convolution, signals (variables) in the node domain are denoted as  $a$  and  $b$ . The definition of graph convolution can

be formulated as follows:

$$a * b = U (U^T b) (U^T a), \quad (5)$$

where  $*$  denotes the convolution operation.

ChebyNet (Defferrard et al. 2016) is proposed to reduce the computational complexity by utilizing the graph convolution definition mentioned earlier (Equation (6)). For the sake of simplicity and computational efficiency, Kipf et al. obtained an approximation for the  $l + 1$  layer network in ChebyNet using a first-order approach (Kipf and Welling 2016):

$$H^{l+1} = \sigma \left( \hat{D}^{-\frac{1}{2}} \hat{A} \hat{D}^{-\frac{1}{2}} H^{(l)} W^{(l)} \right), \quad (6)$$

where  $H$  refers to the characteristics of nodes in the graph, while  $W$  signifies the network parameter subject to learning;  $\hat{A} = A + I_n$ ,  $\hat{D}_{ii} = \sum_j \hat{A}_{ij}$ , and the  $\sigma(\cdot)$  is a nonlinear activation function.

Our proposed GCN model utilizes 2-layer convolutions for each pattern graph, with the activation function being Rectified Linear Unit (ReLU) in each layer convolution. The formula for overall forward propagation can be expressed as follows:

$$f(X, A) = \text{ReLU} \left[ \hat{A} \text{ReLU} \left( \hat{A} X W^{(0)} \right) W^{(1)} \right], \quad (7)$$

where,  $f(X, A) \in \mathbb{R}^{n \times d}$ ,  $X \in \{X_{PC}, X_{SR}, X_{GCM}\}$ ,  $A \in \{A_{PC}, A_{SR}, A_{GCM}\}$  denotes the dimension of the output features. The weight parameter matrix  $W^{(0)}$  represents the learnable weights of the first convolution layer, while  $W^{(1)}$  represents the weight parameter matrix of the second convolution layer.

To make predictions on the category labels of the entire graph, we utilize various GCN models that leverage both the graph structure and node content information from a given subject as inputs. The convolution layer plays a crucial role in generating higher-order representations of the graph. Subsequently, in graph classification tasks, a readout layer is commonly employed to derive a representation at the graph level for the entire graph. Drawing inspiration from Lee et al., we integrate maximum pooling and average pooling operations to aggregate node features and generate fixed-size vectors representing 3 patterns derived from the same subject. This readout layer is defined as follows (Lee et al. 2019):

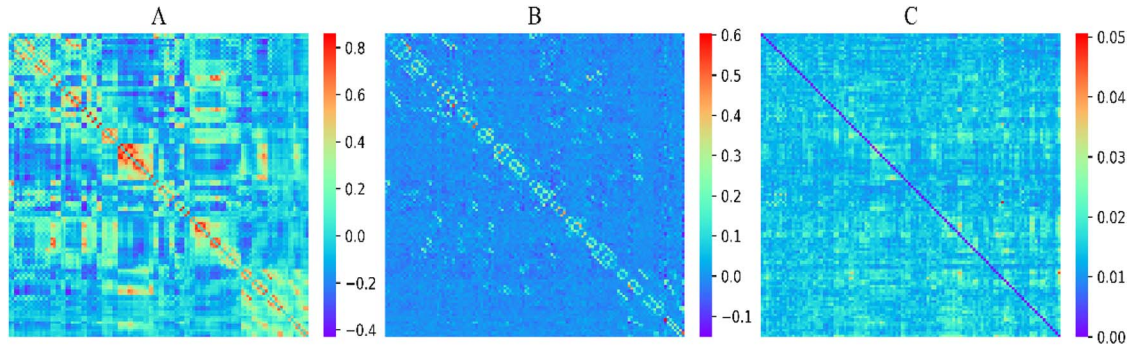
$$F = \frac{1}{n} \sum_{i=1}^N f_i(X, A) \parallel \max_{i=1}^N f_i(X, A), \quad (8)$$

where  $f_i(X, A)$  denotes the  $i$ th ROI acquired via convolution, and  $\parallel$  signifies the merging operation that combines these feature vectors from different ROIs into a single vector.

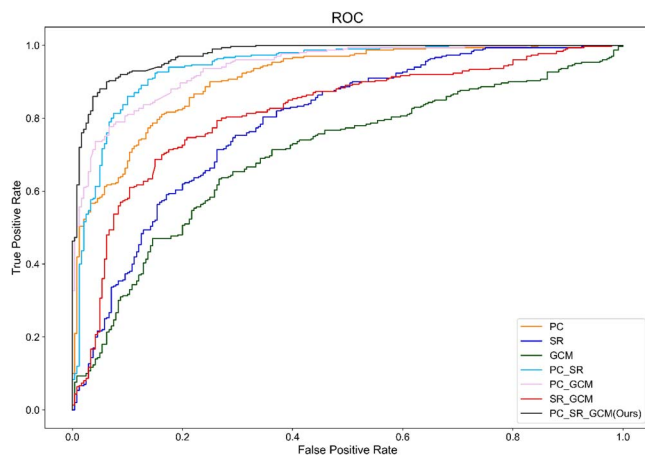
Lastly, we obtain 64D vectors that represent the graphs with various patterns:  $F_k = [F_{1k}^{(64)}, F_{2k}^{(64)}, \dots, F_{Nk}^{(64)}]^T$ ,  $k = 1, 2, 3$ . These representation vectors, denoted as  $F_{m1}$ ,  $F_{m2}$ , and  $F_{m3}$  correspond to the graph representations of various magnitudes for the  $m$ th subject, where  $m$  ranges from 1 to  $N$ .

### Feature fusion and classification across multiple patterns

We apply equal treatment of the graph-level representation vectors obtained from the previous readout operation by merging them. This leads to a 192D novel graph representation vector for each subject, ( $F_1 \parallel F_2 = [F_1^{192}, F_2^{192}, \dots, F_N^{192}]^T$ ), which is then subjected to 3 fully connected layers for further exploration of feature information within the graph. Ultimately, the classification task relies on the output generated by the last fully connected layer.



**Fig. 2.** The connectivity networks derived from the A) PC, B) SR, and C) GCM methods.



**Fig. 3.** ROC curves of 7 methods.

## Implementation details

The implementation of the MPGCN model involves PyTorch and leverages a GPU (NVIDIA GeForce RTX with 16 GB memory) to enhance the computational speed. The MPGCN framework comprises 3 GCN modules, each consisting of 2 convolutional layers and a readout layer. These independent GCN modules are incorporated into the framework, which also consists of 3 fully connected layers. The final layer is utilized for prediction purposes as well. Both the convolutional and fully connected layers employ the ReLU nonlinear activation function. The number of neurons in each graph convolution layer is specified as 32. For the fully connected layers, we have chosen to use 192, 32, and 16 neurons, respectively. To address the issue of overfitting, a dropout rate of 0.5 has been implemented for these layers. The optimization process involves utilizing the Adaptive Moment Estimation (Adam) optimizer (Kingma and Ba 2014) with a learning rate set at 0.025 and a regularization parameter of 0.00001. The model is trained for 150 epochs, and this training process is repeated for 30 times to achieve a better result stability.

## Experiments and results

### Experimental settings

Given the limited sample size, we employed a random selection process to allocate 70% of the samples as training data, 10% as validation data, and the remaining 20% as test data. During each training iteration, we conducted validation and saved the parameters corresponding to the highest validation performance. These saved parameters were then utilized for testing the model.

This process was repeated for a total of 30 times, and average results along with their SDs were recorded for each evaluated method. To ensure fair comparison, identical data partitioning and model training strategies were applied when evaluating our MPGCN approach alongside other competing methods. To assess the efficacy of various approaches, we employed 5 performance evaluation measures: accuracy, recall, precision, F1 score, and area under Receiver Operating Characteristic curve (AUC). These metrics provide a comprehensive evaluation of the model's performance: true positive (TP), true negative (TN), false positive (FP), and false negative (FN) are denoted as follows: TP represents correctly identified positive samples, TN signifies accurately identified negative samples, FP indicates incorrectly identified positive samples, and FN denotes erroneously identified negative samples. The initial 4 measures are described as follows:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FN} + \text{FP} + \text{TN}}, \quad (9)$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad (10)$$

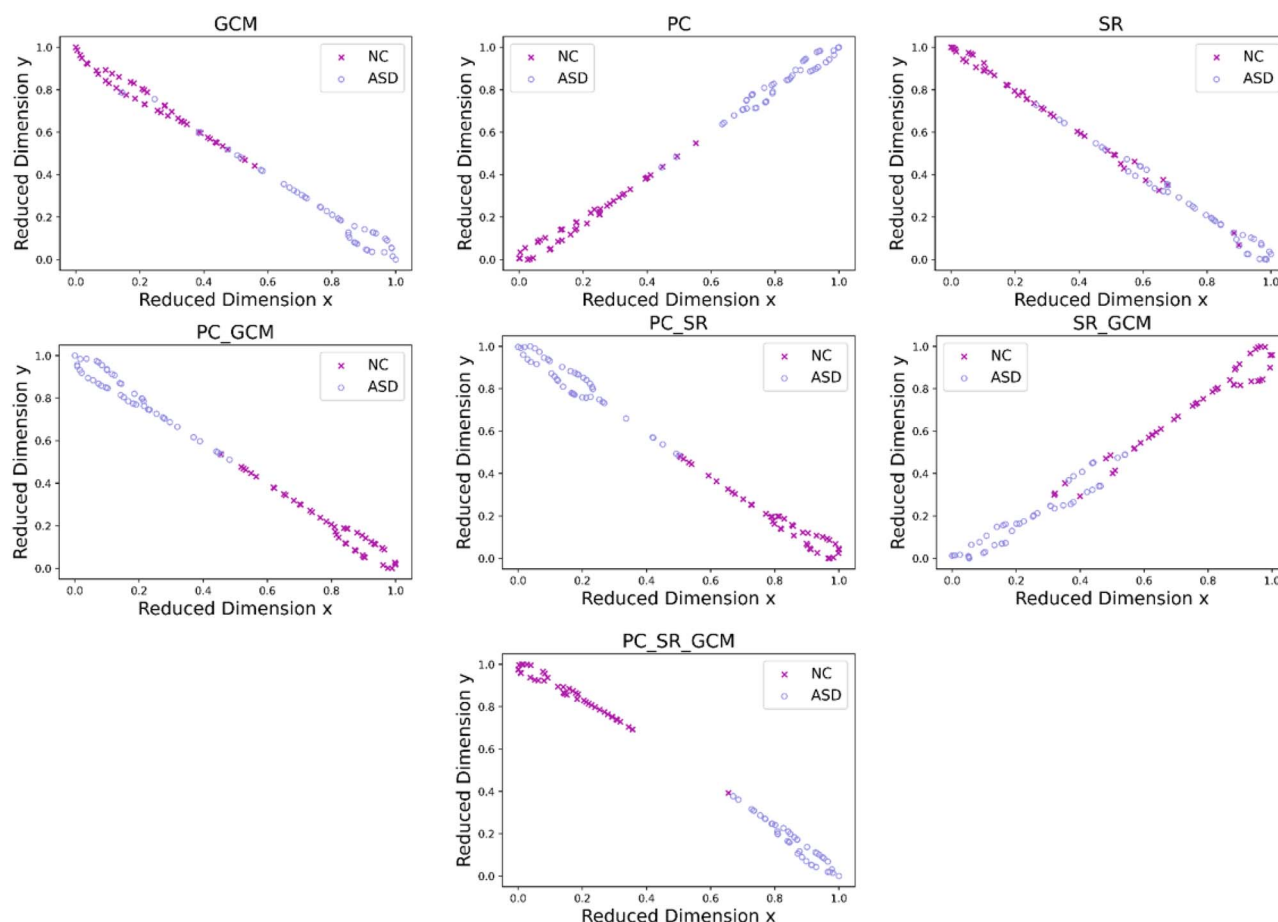
$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad (11)$$

$$\text{F1 score} = \frac{2 * \text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (12)$$

The classification performance of the corresponding model improves as these metrics increase. Additionally, the ROC curve is created using the true positive rate on the y-axis and false positive rate on the x-axis. The AUC represents the likelihood that randomly selected positive cases will be ranked higher than randomly selected negative cases by the classifier. A higher AUC value, closer to 1, indicates a stronger ability of the model to discriminate between cases. We present Fig. 2 using FBN adjacency matrices obtained from PC, SR, and GCM methods. In Fig. 2, different estimation methods demonstrate unique brain connection topologies as they capture distinct statistical information or relationships among ROIs.

### Methods for comparison

We conducted a comparative analysis of our proposed MPGCN approach with different combinations of partial connectivity patterns and individual connectivity patterns, including: (i) constructing brain networks using Pearson's correlation (PC); (ii) constructing brain networks using SR; (iii) constructing brain networks using the Granger causality mapping (GCM); (iv) combining PC and SR for brain network construction; (v) combining PC and GCM for brain network construction; and (vi) combining SR and GCM for brain network construction.



**Fig. 4.** t-SNE visualization applied to 7 connectivity modality methods.

**Table 2.** Performance results of different models in ASD classification.

| Model     | Accuracy (%)       | Precision (%)     | Recall (%)         | F1 score (%)       | AUC (%)           |
|-----------|--------------------|-------------------|--------------------|--------------------|-------------------|
| PC        | 81.7 ± 14.6        | 81.5 ± 20.6       | 85.6 ± 15.2        | 82.1 ± 16.7        | 90.8 ± 2.2        |
| SR        | 74.7 ± 14.3        | 74.1 ± 25.9       | 80.5 ± 16.4        | 74.0 ± 19.5        | 81.7 ± 2.0        |
| GCM       | 67.7 ± 15.8        | 65.6 ± 25.9       | 73.8 ± 19.7        | 66.7 ± 21.5        | 72.8 ± 2.6        |
| PC_SR     | 89.3 ± 11.6        | 89.3 ± 16.3       | <b>92.2 ± 10.4</b> | 89.7 ± 12.5        | 96.3 ± 1.2        |
| PC_GCM    | 86.8 ± 14.2        | 85.9 ± 18.3       | 90.5 ± 12.9        | 87.1 ± 15.1        | 94.6 ± 0.9        |
| SR_GCM    | 78.4 ± 17.6        | 75.5 ± 27.3       | 83.9 ± 19.4        | 76.9 ± 22.7        | 84.3 ± 2.5        |
| PC_SR_GCM | <b>91.1 ± 10.3</b> | <b>93.5 ± 9.3</b> | 91.0 ± 13.6        | <b>91.6 ± 10.5</b> | <b>97.1 ± 1.2</b> |

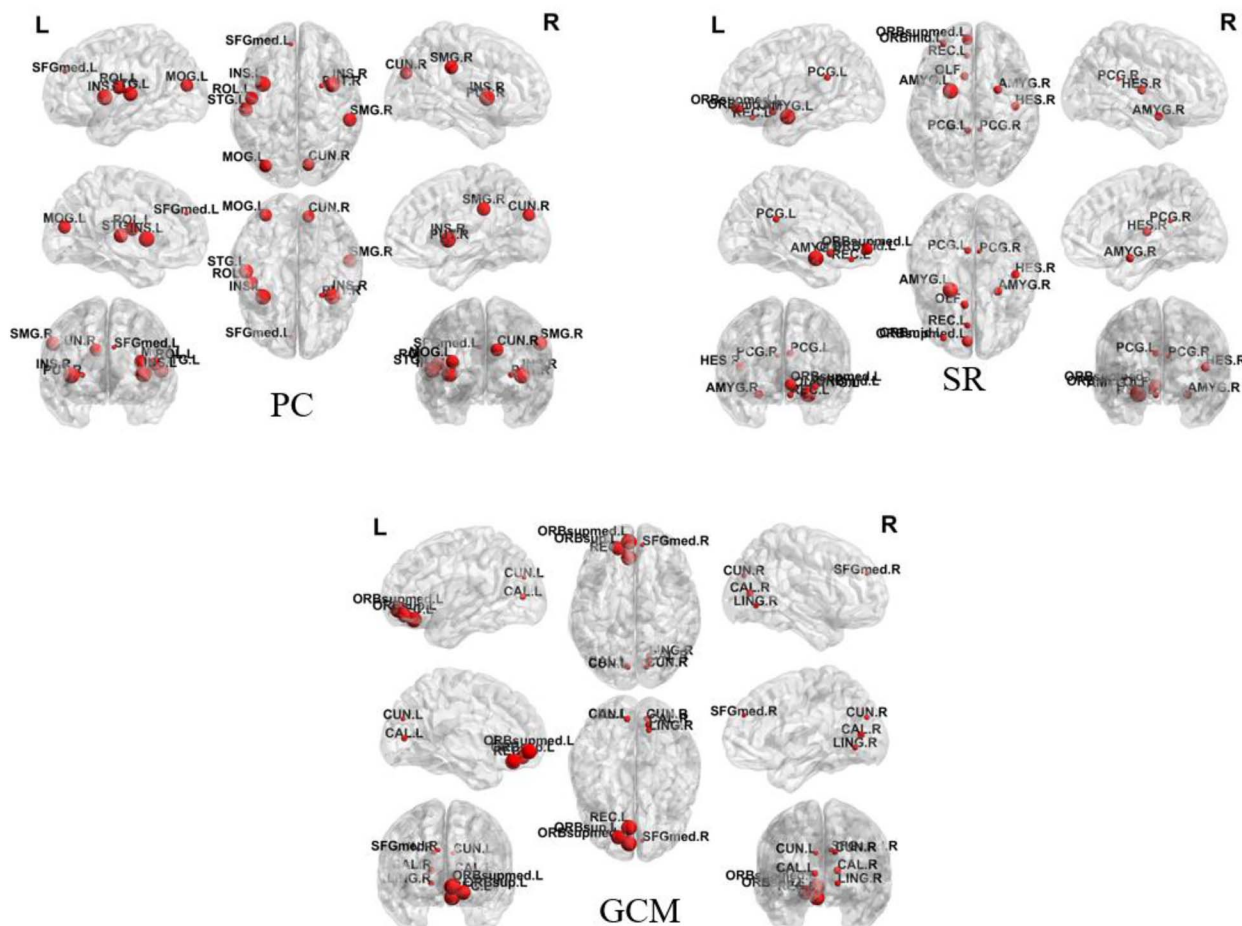
Note that all data in the table are presented as mean ± SD. ASD: autism spectrum disorder. PC: Pearson's correlation; SR: sparse representation; GCM: Granger causality mapping. Boldface denotes the best result in each line.

For the initial 3 categories of single-mode connected brain networks, they comprise a GCN module layer and 3 layers that are fully connected. The GCN module consists of 2 convolutional layers and 1 readout layer (Xu et al. 2018). Both the convolutional and fully connected layers employ nonlinear activation functions for logical operations. The number of neurons in the graph convolutional layers is set to 32 each. In the 3 fully connected layers, there are 64, 32, and 16 neurons, respectively. Furthermore, classification is directly performed by the last layer containing 2 neurons. For the latter 3 types of partially connected brain networks, they have the same parameter settings as the single-mode connected networks. However, compared to the single mode, this method has an additional input. Therefore, it employs an extra GCN module to process the second input. As a result, the first fully connected layer is adjusted to have 128 neurons. In order to ensure a fair comparison, all 6 competing methods utilize a dropout rate of 0.5 for their fully connected layers. The networks are trained using

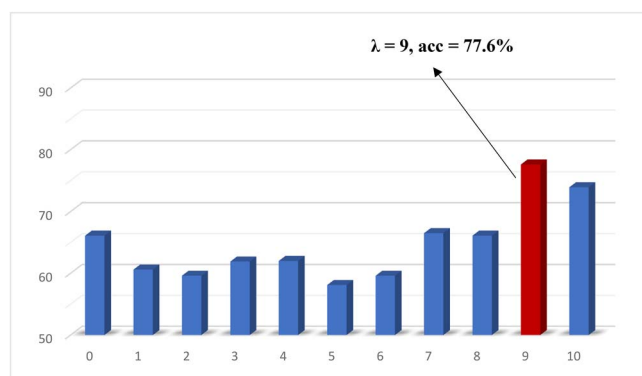
the Adam optimizer and crossentropy loss function. A learning rate of 0.0025 is employed, along with a regularization parameter of 0.00001, and the training process lasts for 150 epochs. This entire procedure is repeated for 30 times, and the mean values as well as SDs of each metric are recorded for analysis purposes.

## Classification results

The results of our MPGCN and the 6 other methods in classifying ASD and NC can be found in Table 2, while Fig. 3 displays the ROC curves for these approaches. Based on the information presented in Table 2 and Fig. 3, several intriguing observations can be made. The use of connectivity information from 2 modalities significantly outperforms using information from a single modality, indicating the effectiveness of integrating different connectivity modalities. Furthermore, the more the connectivity modalities are fused, the higher is the performance. The classifier that fuses 3 different connectivity modalities achieves the best



**Fig. 5.** Relationship between ASD discrimination and brain region contribution.



**Fig. 6.** Sensitivity analysis of different  $\lambda$  for SR.

performance, further demonstrating the effectiveness of MPGCN. It can be observed that combining multiple methods of connectivity modalities leads to more reliable classification results. The results depicted in Fig. 3 demonstrate that as the number of fused connectivity modalities increases, the ROC curve approaches closer to that of the optimal model. The primary cause is that the inclusion of diverse connectivity modalities in FBN characteristics may encompass supplementary data, which can be collectively employed to enhance the classification outcomes.

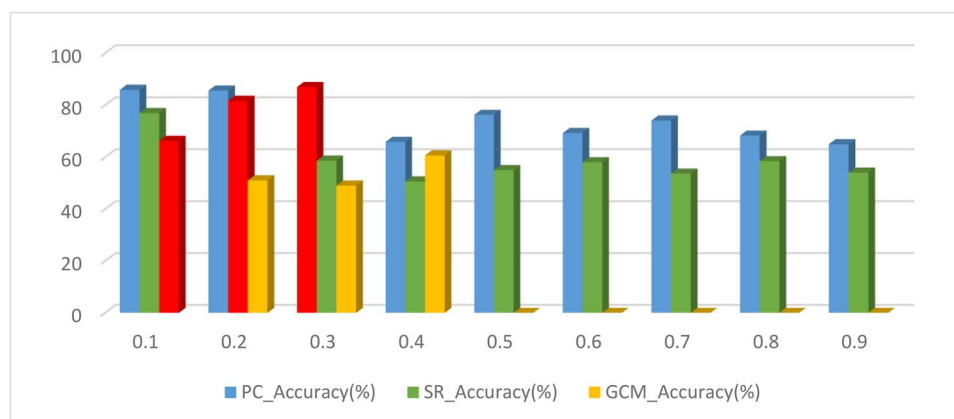
## Visualization of network features

To explore the dispersion of FBN characteristics acquired from various techniques, we employed the t-SNE (Van Maaten and Hinton 2008) algorithm to condense the dimensionality of FBN features obtained from 7 distinct connectivity modalities into a 2D representation. The outcomes are depicted in Fig. 4. It can be observed that when using 2 modalities of information, different types of samples tend to be more separated compared to using a single modality, while samples from the same category tend to be closer to each other. This phenomenon becomes even more pronounced when using 3 modalities of information. It is evident that the more connectivity modalities are fused, the more significant the effect becomes.

To present the connections between the extracted features and brain regions more clearly, we plotted a relationship graph of feature contributions and brain regions under 3 different connectivity patterns in Fig. 5, where the size of the spheres represents the degree of contribution. In Fig. 5, it is interesting to observe that there are varying degrees of brain region contributions corresponding to each pairwise estimation using FCN methods.

## Sensitivity analysis

Please note that the final performance in SR is highly sensitive to the hyperparameter  $\lambda$ . Therefore, in the classification experiments, we further conducted a sensitivity analysis on the classification results with different values of  $\lambda$ . In Fig. 6, we found



**Fig. 7.** Sensitivity analysis of different thresholds for single mode.

that when  $\lambda = 9$ , SR achieved the highest accuracy (77.6%). Consequently, for its implementation, we used  $\lambda = 9$  to construct patients' FCN as the model input.

At the same time, we investigated the relationship between performance and estimated FCN inputs using different thresholds in a single connection mode. In Fig. 7, PC, SR, and GCM achieved the best performance when the threshold values were 0.3, 0.2, and 0.1, respectively, 85.9%, 81.7%, and 66.3%. For different FCN estimation methods, the threshold corresponding to the best performance does not all appear in the minimum value. This means that the brain FCN may contain some noisy/redundant connections that could negatively affect the performance of the model.

## Discussion

According to the classification results presented in Table 2, it can be observed that the PC method exhibits a higher level of accuracy compared to both the SR and GCM methods. Moreover, when combining information from different connection patterns, the performance of this approach is significantly improved as opposed to using a single connection pattern alone. These findings suggest that the proposed MPGCN method has the potential to greatly enhance the diagnostic accuracy for ASD. It should be noted that distinct discriminative information for diagnosis can be provided by different connection patterns. Additionally, surpassing results obtained by considering only 2 connection patterns further validates the superiority of our approach. Figure 4 clearly demonstrates how our proposed MPGCN method outperforms other connectivity patterns in effectively distinguishing between NC and ASD. In Fig. 5, we extracted features from 3 different constructions of the FCN input in our model and visualized them using the BrainNet Viewer tool to gain a clearer understanding of the relationship between the brain region contributions and ASD diagnosis. Subsequently, we conducted a sensitivity analysis. The relationship between the hyperparameter  $\lambda$  and performance in the SR method was discussed, as shown in Fig. 6. In Fig. 7, we explore the relationship between performance using different thresholds under a single connectivity mode and estimate the FCN inputs. In conclusion, integrating multiple connection patterns allow us to elevate the accuracy of ASD diagnosis.

## Conclusion

In this paper, we propose a MPGCN framework based on rs-fMRI for automated diagnosis of ASD. Specifically, we first constructed

3 different brain networks for each subject based on rs-fMRI data. Then, we proposed using multiple GCN modules to learn the multimodal graph representations for each network and fuse them together to integrate multimodal features. The fused features were subsequently propagated through 3 fully connected layers, and the disease diagnosis was accomplished by leveraging the results of the final fully connected layer. We evaluated the effectiveness of our approach, called MPGCN, on rs-fMRI scans from 92 subjects in the ABIDE database. The results demonstrated the effectiveness of MPGCN in FBN feature learning and ASD diagnosis.

## Author contributions

Wenhao Zhou, Mingxiang Sun, and Xiaowen Xu designed the study. Yudi Ruan and Chenhao Sun acquired the MRI data. Weikai Li and Xin Gao analyzed and interpreted the results of the data and drafted the manuscript. All authors contributed equally to this work and approved the final manuscript.

## CRedit authors statement

Wenhao Zhou (Methodology, Writing—original draft), Mingxiang Sun (Writing—original draft), Xiaowen Xu (Data curation, Investigation, Validation), Yudi Ruan (Resources), Chenhao Sun (Visualization, Writing—original draft), Weikai Li (Funding acquisition, Methodology, Writing—review & editing), and Xin Gao (Methodology, Writing—review & editing).

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